

$$1.1) \operatorname{Re}(\langle m, n \rangle) = \frac{1}{4} (\|m+n\|^2 - \|m-n\|^2)$$

$$\begin{aligned} \|m+n\|^2 &= \langle m+n, m+n \rangle = \langle m, m \rangle + \langle m, n \rangle + \langle n, m \rangle + \langle n, n \rangle \\ &= \|m\|^2 + \|n\|^2 + \langle m, n \rangle + \langle n, m \rangle \end{aligned}$$

$$\|m-n\|^2 = \langle m-n, m-n \rangle = \|m\|^2 + \|n\|^2 - \langle m, n \rangle - \langle n, m \rangle$$

$$\|m+n\|^2 - \|m-n\|^2 = 2(\langle m, n \rangle + \langle n, m \rangle)$$

probožme $\langle n, m \rangle = \overline{\langle m, n \rangle}$, dočkaneme $\langle m, n \rangle + \overline{\langle m, n \rangle} = 2 \operatorname{Re}(\langle m, n \rangle)$

$$\|m+n\|^2 - \|m-n\|^2 = 4 \operatorname{Re}(\langle m, n \rangle) \quad | :4$$

$$\operatorname{Re}(\langle m, n \rangle) = \frac{1}{4} (\|m+n\|^2 - \|m-n\|^2)$$

$$\operatorname{Im}(\langle m, n \rangle) = \frac{1}{4} (\|m-in\|^2 - \|m+in\|^2)$$

$$\|m-in\|^2 = \langle m-in, m-in \rangle = \|m\|^2 + \langle m, n \rangle - i \langle n, m \rangle + \|n\|^2$$

$$\|m+in\|^2 = \|m\|^2 - i \langle m, n \rangle + i \langle n, m \rangle + \|n\|^2$$

$$\|m-in\|^2 - \|m+in\|^2 = 2i(\langle m, n \rangle - \langle n, m \rangle) = 2i(\langle m, n \rangle - \overline{\langle m, n \rangle})$$

$$\langle m, n \rangle - \overline{\langle m, n \rangle} = 2i \operatorname{Im}(\langle m, n \rangle)$$

$$2i(2i \operatorname{Im}(\langle m, n \rangle)) = -4 \operatorname{Im}(\langle m, n \rangle)$$

$$\|m-in\|^2 - \|m+in\|^2 = 4 \operatorname{Im}(\langle m, n \rangle) \quad | :4$$

$$\operatorname{Im}(\langle m, n \rangle) = \frac{1}{4} (\|m-in\|^2 - \|m+in\|^2)$$